



Evidence for rodent pollination in *Erica hanekomii* (Ericaceae)

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Pollination by rodents, originally discovered in Cape Proteaceae, is known from a handful of plant lineages and may be more widespread than current data suggest. Here, we demonstrate the occurrence of this mode of pollination in *Erica*, the largest plant genus in the Cape flora of South Africa. The localized endemic *Erica hanekomii* appears to be adapted for pollination by rodents, on account of its low, mat-forming habit with many dull-coloured, pendulous flower heads, large volumes (up to c. 30 µL per inflorescence) of sucrose-dominated nectar, hook-shaped styles and late winter flowering phenology. Rodents trapped in the vicinity of *E. hanekomii* plants had many thousands of *Erica* tetrads in their droppings. Nocturnal laboratory observation showed that *Acomys subspinosus* (Cape spiny mouse) is adept at manipulating *E. hanekomii* inflorescences and lapping nectar without destroying gynoecia. During terrarium experiments, pollen was deposited mainly on the rostrum, but also on the chest area and forelegs of *A. subspinosus*. No birds or insects were seen to visit *E. hanekomii* in the field. Whereas natural fruit set was 73%, inflorescences from which rodents, but not insects, were excluded using mesh cages, had fruit set of just 3.3%. Exclusion of all visitors resulted in zero seed production. Broken anther rings (a sign of flower visitation in *Erica*) were observed in 90.1% of flowers in the field and 92% of flowers exposed to rodents in the laboratory, but in virtually none of the flowers from which rodents were selectively excluded. This first record of a non-flying mammal pollination syndrome in Ericaceae raises interesting new questions about the origins of the rodent pollination system and the selective factors that lead to its evolution. © 2011 The Linnean Society of London, *Botanical Journal of the Linnean Society*, 2011, **166**, 163–170.

ADDITIONAL KEYWORDS: *Acomys subspinosus* – Cape flora – exclusion experiment – nectar – pollination success – sugar composition – small mammal – vertebrate pollination.

INTRODUCTION

Pollination by rodents was discovered in South African *Protea* species during the early 1970s (Rourke & Wiens, 1977; Wiens *et al.*, 1983). The rodent pollination system has subsequently been reported in four other Cape lineages, namely *Massonia* Thunb. ex Houtt. and *Whiteheadia* Harv. (Asparagaceae *sensu* APG III, 2009 or Hyacinthaceae *s.s.*) (Johnson, Pauw & Midgley, 2001; Wester, Stanway & Pauw, 2009), *Colchicum* L. (Colchicaceae) (Kleizen, Midgley & Johnson, 2008) and *Liparia* L. (Fabaceae) (Letten &

Midgley, 2009). Rodents are largely nocturnal and the difficulty in observing their behaviour means that the extent of the rodent pollination system in the Cape flora may still be underestimated.

Erica L. (Ericaceae), the largest plant genus in the Cape flora, is characterized by spectacular floral diversity. Bird, insect and wind pollination systems are known to occur in the genus (Rebelo, Siegfried & Crowe, 1984; Rebelo & Siegfried, 1985; Rebelo, Siegfried & Oliver, 1985), although relatively few detailed studies of pollination biology have been performed for Cape *Erica* species (e.g. Collins, 1983; Geerts & Pauw 2010). One outstanding mystery has been the pollination of *Erica hanekomii* E.G.H.Oliv. (Fig. 1A). Oliver & Oliver (1999) noted small heaps of dissected

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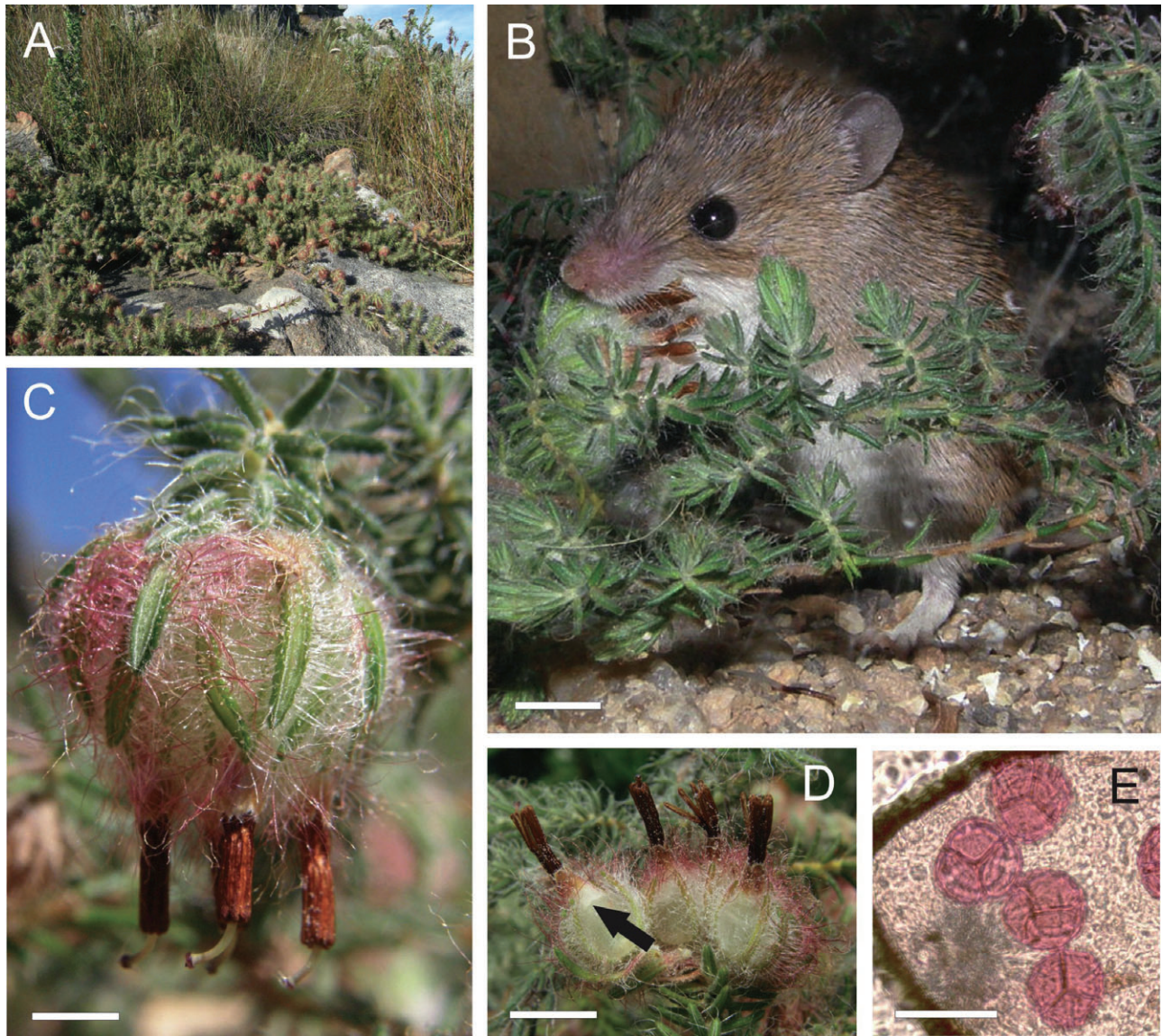


Figure 1. A, *Erica hanekomii* habit and habitat. B, *Acomys subspinosus* foraging on *Erica hanekomii* in terrarium. C, mature *Erica hanekomii* inflorescence with hooked styles. D, bottle-shaped *Erica hanekomii* corollas, inundated with nectar (level indicated by the arrow). E, *Erica* tetrads in droppings of *A. subspinosus*. Scale bars, 5 mm (B); 2 mm (C); 5 mm (D); 50 μ m (E). Photographs by Ross C. Turner.

inflorescences beneath plants of this species and speculated that this could result from the activities of rodents that somehow pollinated flowers in the process of breaking them apart. Although flower damage by rodents has been reported in other systems, legitimate rodent pollinators typically display non-destructive behaviour; for example, Wiens *et al.* (1983), Letten & Midgley (2009) and Biccard & Midgley (2009).

In many respects, flowers of *E. hanekomii* conform to the rodent pollination syndrome, as outlined by Wiens *et al.* (1983). Traits possessed by *E. hanekomii*

that are consistent with the syndrome include the tightly packed inflorescences of up to three whorls of four flowers; stout inflorescence peduncle, apically hooked styles (Fig. 1C), inflorescences close to the ground (< 30 cm, Fig. 1A); dull flower colour (dull green to dull-reddish green, with dull red, glandular and eglandular hairs, Fig. 1C, D); copious production of nectar and winter flowering (peak flowering occurs during August–November to late Winter–early summer) (R. C. Turner, pers. observ., 2001–2010; Oliver & Oliver, 1999). Finally, *E. hanekomii* is a localized endemic that is consistent with the

'restricted distribution hypothesis' of Wiens *et al.* (1983), which was based on the small, highly localized populations, often associated with specific soil types, of virtually all of the rodent-pollinated Proteaceae that they studied.

In this paper, we address the hypothesis that rodents were legitimate pollinators of *E. hanekomii*. We assessed various floral traits, including nectar properties, determined if rodents visit flowers and carry pollen and used selective enclosure experiments to determine the consequences of rodent visits on plant fecundity and to test for the possibility of autonomous self-fertilization.

MATERIAL AND METHODS

STUDY SITE

Field studies were performed within the only known population of *E. hanekomii*: a rocky, montane ridge at the summit of the Middelberg Pass, southern Cederberg, South Africa (−32.604S, 19.138E; 1170 m). Owing to a recent fire in the area (March, 2009), only approximately 100 mature plants were available for study purposes. Through field observation of seedling growth rates, we estimate a minimum age of at least 5 years before plants become reproductively viable. No seedling regeneration was yet evident in recently burnt vegetation. As the pollen tetrads of *E. hanekomii* could potentially be confused with those of other *Erica* spp., we checked the study site for other flowering *Erica* spp.

During 2009, plants of *E. hanekomii* were observed during daytime on 26 August; for 4 days between 28 and 31 September; and for 1 day on 24 October. During 2010, a further 2 days of observation were performed from 6–7 August. Total observation time was at least 40 h. Prior to this study, the first author had observed plants *in situ* on 3–4 July 2000.

SMALL MAMMAL TRAPPING AND OBSERVATION

To assess the local occurrence of rodents and establish, using pollen analysis, whether they visited flowers of *E. hanekomii*, we laid out 27 polyvinyl chloride (PVC) Sherman-type gutter traps under plants of *E. hanekomii*, in the north of the population, on the evening of 28 September 2009, which was the peak flowering period. A further 19 traps were laid out in the south of the population on the evening of 29 September. On the evening of 6 August 2010, we laid out a further 24 PVC gutter traps. Traps were baited with peanut butter and rolled oats. Captured rodents were removed from traps at 07:00 h, identified and released at the precise position of their capture (using a global positioning system; GPS) after nocturnal

laboratory studies. Droppings from four rodents were collected from traps and stored in separate Eppendorf tubes in 70% alcohol for later pollen analysis. This was carried out by crushing approximately 1 mm³ of each dropping sample within an alcohol-filled Eppendorf tube and then applying it to a vortex in order to separate heavier faecal matter from lighter pollen grains. Samples were then stained with three drops of fuchsin dye and then reapplied to the vortex. One slide with coverslip was made for each sample, each with one drop of sample material, and examined at 100× magnification. Slide coverslips (± 22 mm²) were scanned in four rows and the number of *E. hanekomii* pollen tetrads counted per 10 000 µm² at four approximately equidistant points in each row, i.e. 16 sample areas per coverslip. *Erica hanekomii* pollen tetrads are c. 40 µm in diameter.

Pollination behaviour of rodents was observed nocturnally on the evenings of 29 and 30 September 2009 in a 1.0 × 0.5 × 0.5 m), gravel-lined glass terrarium containing multiple flowering branches of *E. hanekomii*, and of a flowering, naturally co-occurring alternative, *Aspalathus cordata* (L.) R. Dahlgren. Rodents were provided with a cardboard cylinder as a shelter, and rolled oats. Two batches of mature inflorescences were introduced to the terrarium each evening and observations took place continuously until all inflorescences had been visited, and for approximately 4 h each evening.

SELECTIVE EXCLUSION EXPERIMENTS

To determine whether rodents are important for seed production in *E. hanekomii* and whether plants are capable of autonomous seed production, we selectively excluded either vertebrates or all flower visitors, respectively. This was carried out on 10 separate mature plants at different localities within the population. As the plants are sprawling, treatments were applied to different inflorescences on each plant. Selective exclusion of vertebrates was achieved by covering branches with wire 'chicken-wire' enclosures (aperture ± 15 × 17 mm) and exclusion of all visitors was achieved by covering inflorescences with bridal-veil enclosures (aperture ± 0.8 mm in diameter). Uncovered inflorescences were used as controls. Treated inflorescences were examined 1 month later to assess rates of visitation and seed set. Visitation to flowers of many *Erica* spp., including *E. hanekomii*, is easily scored by determining if the anther rings have been disturbed (Geerts & Pauw, 2010). Mature, unvisited flowers have an unbroken anther ring, i.e. the anther thecae remain connected laterally to each other, forming a cylinder-like structure through which the apically hooked style is exerted. When manipulated, anther rings break (are triggered) and pollen is

visibly ejected from apical anther pores. Visits by *Acomys subspinosus*, as observed in the laboratory part of this study, typically leave anthers in a state of disarray, occasionally with one or two thecae having been removed to facilitate nectar lapping. Seed set was established from observation of the swelling of ovaries and their subsequent dissection to observe developing seeds, which are distinguishable from ovules by their larger size and 'honey-combed' seed testa.

The effects of selective exclosure on the frequency of inflorescences with flowers that had broken anther rings or set seed were assessed with χ^2 -contingency tests. In these analyses, inflorescences, rather than flowers, were treated as the units of replication as they are tightly congested.

We also assessed predation by insects and rodents upon inflorescences in the field by randomly sampling untreated inflorescences from 10 mature plants at different localities within the population, and then assessing whether or not inflorescences had been removed from plants and dissected; whether or not styles were removed; and whether or not corollas had puncture holes in their bases, typically made by robbing insects, or potentially by an opportunistic rodent as suggested by Oliver & Oliver (1999).

NECTAR PROPERTIES

Nectar from 114 inflorescences (all four flowers per inflorescence), from separate plants, was sampled throughout the population using disposable, calibrated 1-ml³ micro-syringes, accurate to 0.005 mL. Nectar sugar concentrations were measured using an Eclipse hand-held refractometer (Bellingham & Stanley Ltd) capable of taking measurements of up to 50% sugar concentration. Sampling took place between 19:00 h and midnight. Nectar samples were spotted onto four 7.0-cm Whatman filter papers and dried for later determination of constituent free sugars using HPLC methods, as described by Brown, Downs & Johnson (2009). To determine if nectar properties of rodent-pollinated plants show evolutionary convergence, we also assembled data on the nectar properties of Cape plants known to be rodent pollinated.

RESULTS

SMALL MAMMAL TRAPPING AND OBSERVATION

During September 2009, a total of four *A. subspinosus* captures were made: two adults and one male juvenile. These three mice were used for terrarium observation on the evenings of 29 and 30 September 2009. A further captured individual escaped before we could

determine gender or size but left positively identifiable droppings for later pollen analysis. During August 2010 we made a further nine captures: three *A. subspinosus*, two *Elephantulus edwardii* (Cape rock elephant shrew) and four *Aethomys namaquensis* (Namaqua rock mouse).

Droppings of all four *A. subspinosus* mice captured in 2009 contained substantial numbers (thousands) of *E. hanekomii* pollen grains, easily identifiable by their tetrad structure. The density of pollen tetrads was 14.7 per 10 000 μm^2 (range 2–48). In 2010, the mean density of tetrads was 12.1 per 10 000 μm^2 (range 3–35) for droppings from the three *A. subspinosus* individuals, 0.9 tetrads per 10 000 μm^2 (range 0–4) for the two *E. edwardii* individuals and 0.4 tetrads per 10 000 μm^2 (range 0–4) for the four *A. namaquensis* individuals. The only other flowering *Erica* sp. on site during the study was *Erica inaequalis* (Klotzsch) E.G.H.Oliv., a species with small, tubular, pink flowers and tiny nectaries, $\pm 0.2 \times 0.4$ mm (Oliver, 2000). This probably insect-pollinated species was past peak flowering during the study and we could therefore be certain that observed pollen tetrads were those of *E. hanekomii*.

The large number of *E. hanekomii* tetrads present in droppings is ostensibly attributable to *A. subspinosus* individuals fastidiously cleaning and preening their forelegs and facial areas between pollination forays, possibly to remove excess pollen and stickiness acquired from the many hundreds of glandular hairs surrounding each flower and inflorescence.

Upon introduction of flowering branches to the terrarium, mice immediately (within 5 s) exited their hide and began foraging amongst inflorescences (Fig. 1B). Immature inflorescences of *E. hanekomii* and all flowers of *A. cordata* were ignored. During these experiments, 92% of inflorescences had the anther rings in their flowers triggered and none was predated. These levels are almost identical to those observed in the field (90.1% triggered and no predation), indicating high natural visitation rates. *Acomys subspinosus* was adept at manipulating mature inflorescences in order to lap nectar. Mice regularly nibbled away a portion of one of four corolla lobes in order to create a larger aperture for lapping, regularly removing one or two anthers to further facilitate access to the corolla mouth. Styles were never removed. *Acomys subspinosus* induced nectar to rise around the anther filaments and style (nectar-channelling) by firmly squeezing the inflorescence with its forelegs while lapping. Because of the pronounced overbite possessed by these rodents, the rostrum, mouth, upper chest area and forelegs come into contact with anthers and styles of all visited inflorescences.

Table 1. The effects of selective exclusion of animal visitors from inflorescences on rates of pollination and seed set in *Erica hanekomii*

Exclosure type	No. of plants	No. of inflorescences	Inflorescences with flowers with broken anther rings (%)	Inflorescences with flowers that set seed (%)
Open	10	49	76	73.5
Mesh (excludes vertebrates)	10	30	3.3	3.3
Veil (excludes insects and vertebrates)	10	38	8	0

See text for statistical analysis.

SELECTIVE EXCLUSION EXPERIMENTS

Selective exclosure of inflorescences had a highly significant effect on the frequency of inflorescences with flowers that had broken anther rings ($\chi^2 = 22.1$, $P < 0.0001$, Table 1), indicating a strong effect of the exclosures on rates of pollinator visitation. Thus, 76% of open inflorescences (controls) had flowers with broken anther rings, but only a single inflorescence in the chicken-wire (all vertebrates excluded) treatment group had broken anther rings. Three inflorescences within the bridal-veil treatment (all visitors excluded) had broken anther rings. These were probably disturbed when the exclosure was applied and flowers did not set seed.

Selective exclosure of inflorescences had a highly significant effect on the frequency of inflorescences with flowers that set seed ($\chi^2 = 68.3$, $P < 0.0001$, Table 1). Seed set occurred in 73.5% of open inflorescences, but only 3.3% of vertebrate-excluded inflorescences and none of the inflorescences from which all visitors were excluded. There was no evidence, for example, of puncture holes in corollas, of nectar robbing by insects and no visits by flying insects were observed during the study. Thrips were observed within several individual flowers, but owing to their tiny size ($c. < 2 \text{ mm} \times < 0.5 \text{ mm}$) (Picker, Griffiths & Weaving, 2004) are not capable of displacing anther rings of the larger southern African *Erica* spp., especially those of the robust-flowered *E. hanekomii*.

NECTAR PROPERTIES

The mean ($\pm$ SD) nectar volume was $9.0 \pm 7.5 \mu\text{L}$ (range 2.5–30 μL). The mean ($\pm$ SD) nectar sugar concentration was $24.13 \pm 6.6\%$ (range 15.5–47.0%). The nectar comprised sucrose (61.4 $\pm$ 3%), glucose (7.9 $\pm$ 0.5%) and fructose (29.8 $\pm$ 3.5%).

DISCUSSION

The results of this study are consistent with the hypothesis that *E. hanekomii* is rodent pollinated. *Acomys subspinosus* visited flowers in a manner that

was non-destructive to styles and which resulted in contact with anthers and stigmas; flowers from which vertebrates were excluded seldom had anthers triggered and set few or no seeds relative to open-pollinated control; substantial numbers of *E. hanekomii* pollen tetrads were found in *A. subspinosus* droppings over 2 years of trapping; and no other pollinators were observed.

Birds would also have been excluded by the mesh exclosures, but we doubt whether they play a role in pollination of the study species as no birds were observed on the plants during extensive daylight observations. Given the unusual yet purposeful pollination behaviour demonstrated by *A. subspinosus* during terrarium experiments, it may be that pendulous flower heads have evolved to escape the notice of birds and insects, which would probably be ineffective pollinators, rather than as a means of presenting an inflorescence at a favourable angle for rodents to pollinate. *Acomys subspinosus* is adept at manipulating inflorescences and at climbing and jumping (R.C. Turner, J.J. Midgley, pers. observ., this study) and flowers need only be close to the ground and plants sturdy enough for rodents to climb without damaging them.

The low occurrence of pollen tetrads in droppings of *E. edwardii* and *A. namaquensis* relative to those of *A. subspinosus* suggests that no legitimate pollination behaviour, pollen-preening and therefore little pollen ingestion occurred in those two species. Although we cannot discount further potential rodent pollinators, dropping analysis from 2009 and 2010 suggest that only *A. subspinosus* pollinated *E. hanekomii* during our studies.

Several studies have reported a 'yeast-like' or 'yeasty' floral scent as being a component of the rodent-pollination syndrome (Wiens *et al.*, 1983; Johnson *et al.*, 2001; Kleizen *et al.*, 2008; Biccard & Midgley, 2009), although only one recent study analyses this in detail (Johnson *et al.*, 2011). We could discern a faint 'musty' odour in sampled nectar, but no floral scent compounds were detected in preliminary analyses based on the methods detailed by Johnson *et al.* (2011).

The habitat of *E. hanekomii* (a rocky mountain ridge top) corresponds with the known habitat of *A. subspinosus* (Wiens *et al.*, 1983; Fleming & Nicolson, 2002; Letten & Midgley, 2009). Our findings support the idea expressed by Letten & Midgley (2009) that *A. subspinosus* should be considered a 'keystone' species, as it pollinates Ericaceae (this paper), Fabaceae (Letten & Midgley, 2009) and Proteaceae (Wiens *et al.*, 1983; Biccard & Midgley, 2009) and disperses seeds of Proteaceae (Midgley *et al.*, 2002).

Whereas other rodents such as *Rhabdomys pumilio* have been shown to destroy flower heads of *Liparia parva* Vogel ex Walp. (Fabaceae) (Letten & Midgley, 2009) and of *Protea* spp. (Wiens *et al.*, 1983; Biccard & Midgley, 2009), no *R. pumilio* individuals were trapped during 2009 or 2010, and the absence of predated, dissected inflorescences beneath plants suggests that *A. subspinosus* does not destroy inflorescences as *R. pumilio* is known to do. Although only seven *A. subspinosus* individuals were captured for the purposes of this study, we trapped across the *E. hanekomii* population during both years, utilizing most of the available, mature, flowering individuals.

Wiens *et al.* (1983) proposed 'that differences in population structure may be the single most important factor regulating the evolution of pollination by non-flying mammals in *Protea*.' They referred to this as the 'restricted population hypothesis', which in rodent-pollinated Proteaceae involved the shift from bird pollination towards rodent pollination, with localized and ubiquitous rodent pollinators hypothesized to provide a more reliable pollination service for the localized plants than birds, which would focus their foraging activities in larger populations. Thus, the Allee effect, whereby fitness is reduced in small populations, which has been observed in both insect- and bird-pollinated plants (Ward & Johnson, 2005; Lamont, 1993), could be the underlying reason for a shift to pollination by rodents in small populations.

Erica hanekomii is also a range-restricted endemic known only from one rocky ridge ($\pm 0.21 \text{ km}^2$) in the southern Cederberg, Cape Province, South Africa. The total population contains fewer than 500 plants, most of which are juveniles. However, contrary to observations by Wiens *et al.* (1983), *E. hanekomii* does not occur on a specialized substrate, rather one that is replicated throughout the Kouebokkeveld and southern Cederberg: Quartzitic Table Mountain Sandstone with quartzitic and ferrous (manganese) gravels. The locality has been subjected to too-frequent fire over the past 15 years, four separate fires burning all or portions of the population within this time frame (R. C. Turner, pers. observ., 2000–2010; Oliver & Oliver, 1999).

Erica hanekomii is currently classified as a vulnerable (VU) taxon of conservation concern in the 2009 IUCN Red List of Threatened Species (Raimondo *et al.*, 2009). Understanding pollination syndromes is crucial for conservation management of habitats that are increasingly threatened by anthropogenic activities and global change factors and the mutualism demonstrated in this paper warrants a reassessment of the current threat status of *E. hanekomii*. Why rodent-pollinated Cape plant species tend to occur in small, localized populations is unknown (Rebello, 1987). *Acomys* provides an efficient pollination service for *E. hanekomii*, with recorded triggering rates of $\pm 90\%$ in the wild and terrarium and seed set rates of 73.5% in the wild, suggesting that fecundity of *E. hanekomii* is not pollen-limited, in contrast to other Cape plant species (Johnson & Bond, 1997). As the number of known rodent-pollinated Cape plants increases, the extent of convergent evolution in floral traits such as colour and morphology is becoming more evident. It is not yet clear, however, whether there are distinctive nectar traits associated with pollination by rodents. The mean nectar sugar concentration of *E. hanekomii* (c. 24%) falls within the recorded range of known therophilous taxa (Table 2). It does not seem that the nectar of rodent-pollinated plants is dominated by any particular sugars (Table 2). Also of interest is the similarity between *E. hanekomii* nectar sugar composition ratios and that of sunbird-pollinated *Erica* spp., as reported by Barnes, Nicolson & Van Wyk (1995). This suggests that the evolution of rodent pollination from a bird-pollinated ancestor would not require changes in sugar composition, although it is unknown whether or not changes would be required in a syndrome involving evolution of rodent pollination from insect pollination. Our findings and subsequent comparisons agree with Barnes *et al.* (1995), that different pollination syndromes within clades (families or genera) are rarely characterized by differences in nectar properties. Whether this is because of phylogenetic constraints or because pollinators are not choosy remains to be determined in most cases.

Several other *Erica* taxa possess characters suggestive of pollination by rodents and deserve closer study, in particular *E. eriophoros* Guthrie & Bolus, *E. involucrata* Klotzsch ex Benth., *E. senilis* Klotzsch ex Benth., *E. tegetiformis* E.G.H. Oliver and *Erica pudens* H.A. Baker. However, none possess these characters as comprehensively as *E. hanekomii*, nor has any been reported to produce nectar in comparable quantities. It is therefore possible that specialized rodent pollination has only evolved once in *Erica* or was once more common and is now represented by one or a handful of extant species.

Table 2. Nectar properties of rodent-pollinated plants in the Cape flora

Taxon	Pollinator	Nectar sugar (%) $\bar{x} \pm SD$	Nectar volume (μ L) $\bar{x} \pm SD$	Sucrose (%)	Glucose (%)	Fructose	Xylose	Reference
<i>Erica hancokii</i>	<i>Acomys subspinosus</i>	24 $\pm$ 6.59	9.0 $\pm$ 7.5 [†]	61.4 $\pm$ 3	7.9 $\pm$ 0.5	29.8 $\pm$ 3.5	0	This study
<i>Liparia parva</i> subsp. <i>parva</i>	<i>Acomys subspinosus</i>	19.7 $\pm$ 5.1	10.7 $\pm$ 6.2*	41.7	26.7	31.7	0	Van Wyk (1993); Letten & Midgley (2009)
<i>Protea humiflora</i>	<i>Acomys subspinosus</i> ; <i>Aethomys namaquensis</i>	37.8 $\pm$ 6.9	8.8*	48.3	14.5	30.5	6.75	Wiens <i>et al.</i> (1983); Nicolson & Van Wyk (1998)
<i>Protea nana</i>	<i>Myomyscus verrauxii</i> ; <i>Aethomys namaquensis</i>	29.4 $\pm$ 6.2	> 1000 [†]	64	5	30	1	Nicolson & Van Wyk (1998)
<i>Protea cryophila</i>	<i>Acomys subspinosus</i>	33.3 $\pm$ 7.2	Not reported	27.6	34.3	33.3	0	Wiens <i>et al.</i> (1983)
<i>Massonia depressa</i>	<i>Gerbillurus pabea</i> ; <i>Aethomys namaquensis</i>	$\pm$ 20	17.7 $\pm$ 26*	42	23	32	3	Johnson <i>et al.</i> (2001)
<i>Whiteheadia bifolia</i>	<i>Aethomys namaquensis</i>	27 $\pm$ 13.9	7.9 $\pm$ 5.9*	7	27	66	0	Wester <i>et al.</i> (2009)

*Amount per flower.

†Amount per inflorescence.

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